Triangle Centers, Ceva, Menelaus & Stewart Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Triangle Centers, Ceva, Menelaus & Stewart Theorem Practice Exercises

1 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{1}{2}$ and $\frac{CE}{EA} = \frac{1}{3}$, find $\frac{AF}{FB}$.

a. $\frac{1}{6}$

b. $\frac{3}{2}$

c. 6 ✓

d. $\frac{5}{2}$

By Ceva's Theorem on concurrent cevians AD, BE, CF, $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{AF}{FB} \cdot \frac{1}{2} \cdot \frac{1}{3} = 1 \Rightarrow \frac{AF}{FB} = 6$.

2 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{2}{3}$ and $\frac{CE}{EA} = \frac{1}{4}$, find $\frac{AF}{FB}$.

a. $\frac{1}{6}$

b. 6 ✓

c. $\frac{8}{3}$

d. $\frac{7}{3}$

By Ceva's Theorem on concurrent cevians AD, BE, CF, $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{AF}{FB} \cdot \frac{2}{3} \cdot \frac{1}{4} = 1 \Rightarrow \frac{AF}{FB} = 6$.

3 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{3}{4}$ and $\frac{CE}{EA} = \frac{2}{5}$, find $\frac{AF}{FB}$.

a. $\frac{10}{3}$ ✓

b. $\frac{3}{10}$

c. $\frac{15}{8}$

d. $\frac{9}{5}$

By Ceva's Theorem on concurrent cevians AD, BE, CF, $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{AF}{FB} \cdot \frac{3}{4} \cdot \frac{2}{5} = 1 \Rightarrow \frac{AF}{FB} = \frac{10}{3}$.

4 In ΔABC shown, points $D \in BC$, $E \in CA$, and $F \in AB$ are chosen so that the cevians AD, BE, CF concur at an interior point P. Given $\frac{BD}{DC} = \frac{1}{4}$ and $\frac{CE}{EA} = \frac{3}{2}$, find $\frac{AF}{FB}$.

a. $\frac{3}{8}$

b. $\frac{1}{6}$

c. $\frac{3}{2}$

d. $\frac{8}{3}$ ✓

By Ceva's Theorem on concurrent cevians AD, BE, CF, $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{AF}{FB} \cdot \frac{1}{4} \cdot \frac{3}{2} = 1 \Rightarrow \frac{AF}{FB} = \frac{8}{3}$.

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5 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 2$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{2}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{3}$

b. $\frac{5}{2}$

c. $\frac{2}{5}$ ✓

d. $\frac{3}{5}$

First note the full exterior distance $BD = BC + CD = 3 + 2 = 5$ (using BC instead of BD is the classic Menelaus error). Applying Menelaus's Theorem to ΔABC and transversal $F - E - D$ gives $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{2}{2} \cdot \frac{5}{2} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{CE}{EA} = \frac{2}{5}$.

6 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 3$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{3}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$ ✓

c. 3

d. $\frac{1}{3} + 1$

First note the full exterior distance $BD = BC + CD = 3 + 3 = 6$ (using BC instead of BD is the classic Menelaus error). Applying Menelaus's Theorem to ΔABC and transversal $F - E - D$ gives $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{3}{2} \cdot \frac{6}{3} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{CE}{EA} = \frac{1}{3}$.

7 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 3$ and $CD = 4$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{4}{2}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{2}{7}$ ✓

b. $\frac{2}{3}$

c. $\frac{7}{2}$

d. $\frac{3}{14}$

First note the full exterior distance $BD = BC + CD = 3 + 4 = 7$ (using BC instead of BD is the classic Menelaus error). Applying Menelaus's Theorem to ΔABC and transversal $F - E - D$ gives $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{4}{2} \cdot \frac{7}{4} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{CE}{EA} = \frac{2}{7}$.

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8 Point D lies on the extension of side BC of ΔABC beyond C such that $BC = 6$ and $CD = 2$. A line through D intersects segment AB at F with $\frac{AF}{FB} = \frac{5}{3}$ and intersects segment AC at E. By Menelaus's Theorem, find $\frac{CE}{EA}$.

a. $\frac{1}{5}$

b. $\frac{20}{3}$

c. $\frac{9}{20}$

d. $\frac{3}{20}$ ✓

First note the full exterior distance $BD = BC + CD = 6 + 2 = 8$ (using BC instead of BD is the classic Menelaus error). Applying Menelaus's Theorem to ΔABC and transversal $F - E - D$ gives $\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{5}{3} \cdot \frac{8}{2} \cdot \frac{CE}{EA} = 1 \Rightarrow \frac{CE}{EA} = \frac{3}{20}$.

9 In ΔABC, the circumcircle (O) has radius $R = 5$ and the incircle (I) has radius $r = 2$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 9

b. 15

c. 5 ✓

d. 6

By Euler's Triangle Theorem (proved by Power of Point I with respect to (O) and Fact 5 on the arc midpoint), $OI^2 = R(R - 2r) = 5(5 - 2 \cdot 2) = 5 \cdot 1 = 5$.

10 In ΔABC, the circumcircle (O) has radius $R = 8$ and the incircle (I) has radius $r = 3$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 25

b. 16 ✓

c. 40

d. 17

By Euler's Triangle Theorem (proved by Power of Point I with respect to (O) and Fact 5 on the arc midpoint), $OI^2 = R(R - 2r) = 8(8 - 2 \cdot 3) = 8 \cdot 2 = 16$.

11 In ΔABC, the circumcircle (O) has radius $R = 11$ and the incircle (I) has radius $r = 4$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 33 ✓

b. 49

c. 77

d. 34

By Euler's Triangle Theorem (proved by Power of Point I with respect to (O) and Fact 5 on the arc midpoint), $OI^2 = R(R - 2r) = 11(11 - 2 \cdot 4) = 11 \cdot 3 = 33$.

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12 In ΔABC, the circumcircle (O) has radius $R = 14$ and the incircle (I) has radius $r = 5$. By Euler's Theorem in geometry, what is the exact squared distance OI^2 between the circumcenter and incenter?

a. 81

b. 126

c. 57

d. 56 ✓

By Euler's Triangle Theorem (proved by Power of Point *I* with respect to (O) and Fact 5 on the arc midpoint), $OI^2 = R(R - 2r) = 14(14 - 2 \cdot 5) = 14 \cdot 4 = 56$.

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